Insurance Purchase Prediction Web Application

Project Overview

The Insurance Purchase Prediction Web Application is a machine learning project that predicts whether a customer is likely to purchase insurance based on their age, annual income, and existing insurance policy status.

The application uses a **Logistic Regression** classification model and provides a user-friendly web interface developed using HTML, CSS, and JavaScript. The backend is built with FastAPI to process user inputs and return prediction results.

Objectives

- Build a machine learning model using Logistic Regression.
- Generate and prepare a customer insurance dataset.
- Evaluate model performance using classification metrics.
- Save the trained model as model.pkl.
- Develop a REST API using FastAPI.
- Create an interactive web interface using HTML, CSS, and JavaScript.
- Validate customer inputs and display prediction results.
- Run and test the complete application using Google Colab.

Technologies Used

- **Programming Language:** Python
- **Machine Learning:** Scikit-learn
- **Data Processing:** Pandas, NumPy
- **Model Serialization:** Joblib
- **Backend:** FastAPI, Uvicorn
- **Frontend:** HTML5, CSS3, JavaScript
- **Development Environment:** Google Colab

Dataset Description

A synthetic dataset containing 1,500 customer records is generated for this project.

Feature	Description

Age	Customer's age
AnnualIncome	Customer's annual income
HasExistingPolicy	Whether the customer already has an insurance policy
Purchased	Target variable indicating insurance purchase (0 or 1)

Note: The dataset is synthetically generated for educational and demonstration purposes. It does not represent actual customer data.

Machine Learning Model

The project uses Logistic Regression, a supervised machine learning algorithm suitable for binary classification.

Workflow

1. Generate the dataset.
2. Separate input features and the target variable.
3. Split the data into training and testing sets.
4. Apply Standard Scaling.
5. Train the Logistic Regression model.
6. Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.
7. Save the trained pipeline as model.pkl.

Application Features

- User-friendly insurance prediction form.
- Customer age and annual income inputs.
- Existing insurance policy selection.
- Input validation on the frontend and backend.
- Prediction of insurance purchase likelihood.
- Estimated purchase probability displayed as a percentage.
- FastAPI backend with interactive API documentation.
- Responsive and clean user interface.

Project Structure

```
insurance-purchase-prediction/
│
├── main.py
├── model.pkl
├── insurance_data.csv
│
├── templates/
│   └── index.html
│
└── static/
    ├── style.css
    └── script.js
```

How to Run the Project

Step 1: Open Google Colab

Open Google Colab and create a new notebook.

Step 2: Install Dependencies

pip install fastapi uvicorn scikit-learn pandas numpy joblib

Step 3: Generate the Dataset and Train the Model

Run the notebook cells to generate the synthetic dataset, train the Logistic Regression model, evaluate its performance, and save the trained pipeline as model.pkl.

Step 4: Create Application Files

Create the backend file main.py and the frontend files inside the templates and static directories.

Step 5: Start the FastAPI Server

python -m uvicorn main:app --host 0.0.0.0 --port 8000

In Google Colab, use the Colab port proxy to access the application from your browser.

Step 6: Test the Application

Enter customer details into the web form and click **Predict Purchase** to view the prediction and estimated probability.

API documentation is available at the /docs endpoint while the server is running.

Model Evaluation

The model is evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The actual evaluation results should be recorded after executing the notebook.

Expected Output

The web application displays one of two results:

- **Likely to purchase insurance**
- **Unlikely to purchase insurance**

It also displays the model-estimated purchase probability as a percentage.

Future Enhancements

- Train and evaluate the model using a real, suitable insurance dataset.
- Improve model performance through feature engineering and hyperparameter tuning.
- Add prediction history and visual analytics.
- Deploy the application to a cloud hosting platform.
- Improve the user interface and accessibility.

Limitations

This project uses synthetic data for educational purposes. Its predictions are not validated for real-world insurance decisions. The estimated probability reflects the trained model and should not be interpreted as a guarantee of customer behavior.

Conclusion

This project demonstrates an end-to-end machine learning application by combining Logistic Regression, data preprocessing, model evaluation, FastAPI, and a web interface. It provides practical experience in integrating a trained machine learning model with a web application.

Project: Insurance Purchase Prediction Web Application
Assignment: Assignment 4
Environment: Google Colab
Purpose: Educational and academic demonstration